Lecture 15

Tuesday, 7 March 2023 10:40 AM

Project topic: HER on Pt(111) → aq. acidic solution

Volcano plots for heterogenous catalysis

ex: 1) $A_2(g) + 2* \rightleftharpoons 2A^*$ appropriate if θ_B is small ; If $\frac{kP_B}{P^\circ} << 1$ → we can assume θ_B is small

2) $A^* + B(g) \rightleftharpoons AB(g) + *$

$B(g) + * \rightarrow B^*$

$\theta^* = \frac{1}{1 + kP_B + \dots}$

$r = r_1 = k_1 P_{A_2} \theta_*^2 - k_{-1} \theta_A'^2$

$K_2 = \frac{\theta_* P_{AB}'}{\theta_A' P_B'}$; $r = k_1 P_{A_2} \theta_*^2 - \frac{k_{-1}}{K_2^2} \left(\frac{P_{AB}'}{P_B'}\right)^2 \theta_*^2$

$= k_1 P_{A_2} \theta_*^2 \left(1 - \frac{k_{-1}}{k_1 K_2^2} \frac{P_{AB}'^2}{P_{A_2} P_B'^2}\right)$

$= P_0/K_1$

$r = k_1 P_{A_2} \theta_*^2 \left(1 - \frac{1}{K_1 K_2^2} \frac{P_{AB}'^2}{P_B'^2 P_{A_2}'}\right)$

γ = approach to equilibrium (set by conditions in reactor)

Site balance:

$\theta_* + \theta_A' = 1$

$\theta_* + \theta_* \frac{P_{AB}'}{K_2 P_B'} = 1$

Vacant sites

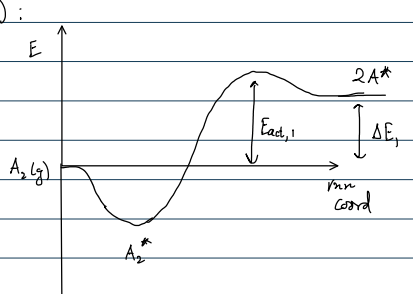
$$r = \frac{k_1 P_{A_2}' (1-\gamma)}{\left(1 + \sqrt{k_1 P_{A_2}'}\,\gamma\right)^2}$$

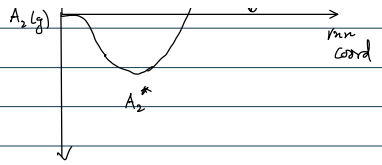
1^* ← ignoring difference b/w H and E

$K_1 = e^{\Delta S_1^\circ / k_B} e^{-\Delta E_1 / k_B T}$

$k_1 = \frac{k_{1o}}{P^\circ} e^{-\frac{\Delta E_{act,1}}{k_B T}}$ } Find k_{1o} from TST

Energy diagram for (1):





If rxn follows a BEP relation: $E_{act,1} = \alpha_1 \Delta E_1 + B_1$

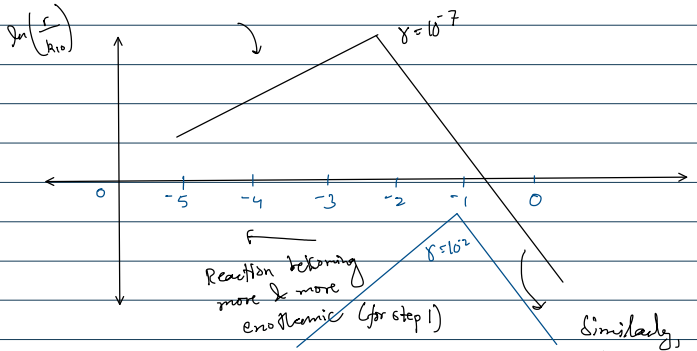
$$\ln\left(\frac{r}{k_{10}}\right) = -\frac{\alpha_1 \Delta E_1 + B_1}{k_B T} + \ln\left(\frac{P_{A_2}}{P^\circ}\right) + \ln(1-\theta) - 2\ln\left[1+\sqrt{e^{\frac{\Delta S_1^\circ}{k_B}} e^{\frac{-\Delta E_1}{k_B T}} \frac{P_{A_2}}{P_0}\,\theta}\right]$$

With the RPBE functional,

$\alpha_1 = 0.9$, $B_1 = 1.34\ eV$; $\frac{P_{A_2}}{P^\circ} \approx 25$

$\Delta S_1^\circ = -200\ \frac{J}{mol\,K}$

} We assume everything else is independent of catalyst surface, i.e. variations in catalyst are given by ΔE_1



(Rate decreases)

↓

Small no. of vacant sites (A is bounding very strongly)

↳ θ_A approaches 1

Similarly, from BEP relations, $E_{act} \propto \Delta E_1$

(The activation energy increases

1) There are two physical processes that limit rate (even w/ 1 RLS)

2) Rate decreases as $\gamma \to 1$

3) Volcano maximum moves to higher ΔE_1 } $r = \frac{k_1 P_{A_2}}{\left(1 + K_1 \frac{P_{A_2}}{P^\circ}\gamma\right)^2}(1-\gamma)$

As $\gamma \to 1$, more A^* on surface so catalyst compensates by weaker binding of A^*

4) Decrease of P_{A_2} shifts max to stronger binding of A^* (to compensate for reduced θ_A)

5) If TT, max. moves to stronger binding (same idea)

5) If T↑, max. moves to stronger binding (same idea)
(Langmuir isotherm → decrease θ_A on surface)

For a given catalyst, it will have just one E_r → the volcano plot actually tells you which one/catalyst would be optimal at the new T, P.

} Also, assuming that the RLS is the same across these changing conditions.

Ex 2: Same idea, but assume θ_{A_2} is not small

1) $A_2(g) + * \rightleftharpoons A_2^*$

2) $A_2^* + * \rightleftharpoons 2A^*$ RLS

3) $A^* + B(g) \rightleftharpoons AB(g) + *$

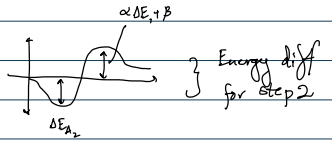
$$r = r_2 = k_2 \theta_{A_2} \theta_* - k_{-2} \theta_A^2 \quad ; \quad K_1 = \frac{\theta_{A_2}}{P'_{A_2} \theta_*} \quad ; \quad K_3 = \frac{P'_{AB} \theta_*}{P'_B \theta_A}$$

$$\theta_A = \frac{1}{1 + K_1 P'_{A_2} + \frac{P_{AB}}{P_B} \frac{1}{K_3}}$$

(*) (A₂*) (A*)

$$r = \frac{k_2 K_1 \frac{P_{A_2}}{P^\circ}\left(1-\gamma\right)}{\left(1 + K_1 \frac{P_{A_2}}{P^\circ} + \sqrt{K_1 k_2 \frac{P_{A_2}}{P^\circ}\gamma}\right)^2} \quad ; \quad \gamma = \frac{1}{K_1 K_2 K_3^2}\left(\frac{P_{AB}^2}{P_B^2 \, P_{A_2}/P^\circ}\right)$$

$$K_1 = e^{\frac{\Delta S_1^\circ}{k_B}} e^{\frac{-\Delta E_{A_1}}{k_B T}}$$



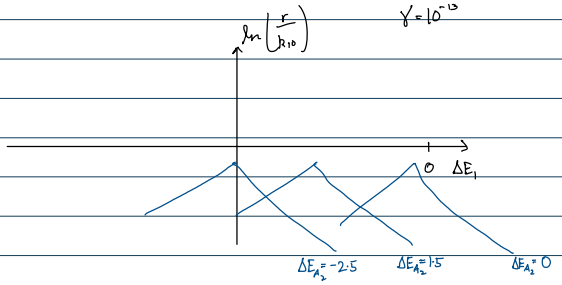
} Energy diff for step 2

$$K_2 = e^{\circ} e^{\frac{-(-\Delta E_{A_2} + \Delta E_1)}{k_B T}}$$

$$k_2 = k_{20} e^{-(\alpha \Delta E_1 + \beta - \Delta E_{A_2})/k_B T}$$

$$\ln\left(\frac{r}{k_{20}}\right) = -\frac{\alpha_1 \Delta E_1 + \beta_1}{k_B T} + \frac{\Delta S_1^\circ}{k_B} + \ln\left(\frac{P_{A_2}}{P^\circ}\right) + \ln(1-\gamma) - 2\ln\left[1 + K_1 \frac{P_{AB}}{P^\circ} + \sqrt{K_1 K_2 \frac{P_{A_2}}{P^\circ}\gamma}\right]$$

With some parameters,



↳ Why does max shift as ΔE_{12} increases?

↳ What happens if ΔE_1 & ΔE_{12} are correlated?